Five Myths about "Kanji" and "Kanji" Learning

Author(s): Yoshiko Mori

Source: *Japanese Language and Literature*, April 2012, Vol. 46, No. 1 (April 2012), pp. 143-169

Published by: American Association of Teachers of Japanese

Stable URL: https://www.jstor.org/stable/41442049

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at https://about.jstor.org/terms

American Association of Teachers of Japanese is collaborating with JSTOR to digitize, preserve and extend access to *Japanese Language and Literature*

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

Five Myths about *Kanji* and *Kanji* Learning[1]

Yoshiko Mori

ABSTRACT

Many learners of Japanese as a second/foreign language (L2) consider *kanji* one of the most challenging aspects of Japanese learning. This article examines the validity of five common views on *kanji* learning that L2 Japanese learners often have: 1. There are too many *kanji* characters to learn; 2. *Kanji* makes Japanese reading difficult; 3. It is easy to guess the exact meaning of an unfamiliar *kanji* word because its elements provide sufficient information about the *kanji*'s meaning; 4. Because the primary function of *kanji* is to represent meaning, sound plays an unimportant role in *kanji* recognition; and 5. Rote memorization is the only way to learn *kanji*. Research findings suggest that these are actually myths, and that positive attitudes toward *kanji,* and constructive approaches to specific learning tasks increase the chances for individuals to become strategic *kanji* leaners.

1. Introduction

Many students of Japanese as a second/foreign language (L2), especially those from a non-*kanji* orthographic background consider *kanji* one of the most challenging aspects of Japanese learning (Toyoda 1995, 1998; Okita 1997; Mori 1999a; Yamashita and Maru 2000; Gamage 2003; Mori and Shimizu 2007). The challenges in *kanji* learning include difficulty in retention, multiple readings of a single character, visual complexity and similarity, the abundance of polysemous words, and the large number of characters to learn (Toyoda 1995). Such perception of difficulty is, at least partially, attributable to the typological differences between logographic and alphabetic orthographies (Tollini 1994).

Individual students define *kanji* learning tasks differently and reflect upon their own learning from various perspectives (Mori and Shimizu 2007). In addition, perceptions of learners can affect their *kanji* learning behaviors (Mori 1999b, 2002, Mori *et al.* 2007). Consequently, positive attitudes toward *kanji* and constructive approaches to specific *kanji*

Japanese Language and Literature **46** (2012) 143–169
© 2012 Yoshiko Mori

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

learning tasks should be promoted in order to increase the chances for individuals to become strategic *kanji* leaners.

The purpose of this article is to critically examine the validity of beliefs about *kanji* often held by L2 Japanese students, and explain how a better understanding of the orthographic features of *kanji* and the nature of *kanji* learning may enable students to rid themselves of unwarranted preconceptions. The following five beliefs were identified in a series of *kanji* perception studies conducted by the author and her colleagues (Mori 1999a, 1999b; Mori *et al.* 2007; Mori and Shimizu 2007).

1. There are too many *kanji* characters to learn.
2. *Kanji* makes Japanese reading difficult.
3. It is easy to guess the exact meaning of an unfamiliar *kanji* word because its elements provide sufficient information about the word's meaning.
4. Because the primary function of *kanji* is to represent meaning, sound plays an unimportant role in *kanji* recognition.
5. Rote memorization is the only way to learn *kanji*.

It is hoped that a critical review of these beliefs will enable both students and educators to reflect upon their own preconceptions from different perspectives.

2. Myth 1: There Are Too Many *Kanji* Characters to Learn

Novice students may feel that there are an impossible number of *kanji* characters to learn. The number of individual characters commonly used in printed materials, however, is not as high as one would expect. According to a survey conducted by the National Language Research Institute (1963), 200 high-frequency *kanji* characters cover approximately 50% of the characters used in Japanese newspapers and periodicals. Five hundred high-frequency characters cover 75–80%, 1,000 characters 90%, and 2,000 characters 98%. An additional 1000 characters increase the coverage only by 1%. Another survey of the National Language Research Institute (1971) revealed that Sino-Japanese words (i.e., words of Chinese origin) accounted for 50–65% of the total word count in written Japanese, whereas Japanese native words accounted for 26–44%. Since native words are often written in *kanji* or in a combination of *kanji* and *hiragana*, the total proportion of *kanji* words in printed Japanese would be higher than 65%. In addition, the bulk of

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

crucial written vocabulary consists of *kanji* compounds—words consisting of two or more *kanji* characters (Kaiho 1984, Shibatani 1990). These statistics suggest that the relationship between the number of characters and their coverage is not straightforward, and that a large portion of crucial information in printed materials is conveyed by a relatively limited number of high-frequency characters.

This line of thought leads to the pedagogical implications that learners of Japanese, including native children, should first learn a manageable number of high-frequency characters. The learning of 200 basic characters, for instance, enables students to recognize over a half of *kanji* used in published materials, and 500 characters cover three quarters. Upon successfully learning 1,006 *kyōiku kanji* (the characters taught in the six years of elementary school education), L2 readers should be able to recognize over 90% of the characters appearing in authentic materials, which is substantial. This idea also supports the prevailing assumption that Chinese native speakers who have already acquired a good working character knowledge in their first language (L1), possess an advantage in learning to read in Japanese, compared to those from non-*kanji* language backgrounds.

A *kanji* character is often considered a "word" (Atsuji 1993, Packard 2000), or, more precisely, a "morpheme"—the smallest meaningful unit (Joyce 2011). If *kanji* learning involves not only the learning of their orthographic features but also the reorganization of the mental lexicon, then learning 500–1,000, or even 2,000, "words" is not unique to L2 Japanese learning. L2 learners, regardless of the language, must acquire a certain number of basic words in order to function in the target language. L2 English readers, for instance, must acquire 3,000–5,000 words to build a base for comprehension (Nation and Waring 1997). Likewise, the learning of 500–1,000 basic characters or "words" builds a foundation for comprehension in L2 written Japanese.

Needless to say, knowing individual characters does not guarantee that the learner can tell the meaning of *kanji* compounds. As discussed later, the meanings of individual characters often only partially contribute to the meaning of the entire word; e.g., 外見 'outside' + 'to see' = 'physical appearance.' Yet many compounds, even extremely low-frequency technical terms that are exclusively used in some fields, are often composed of high-frequency characters that are used on a daily basis; e.g., 長頭 'dolichocephalic' (in Halpern 1990). Although these may not be always accurate, their meanings are inferable to readers who

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

encounter them for the first time (Halpern 1990, T. Suzuki 1990, Hatano *et al.* 1997). Many English "big" words, in contrast, often consist of low-frequency morphemes of Greek or Latin origin that are rarely used on a daily basis, and their meanings are not transparent (e.g., dolichocephalic).

It is certainly challenging to learn the detailed orthographic features of 500–1,000 characters, many of which are visually far more complex than *kana* or alphabet letters, but there is no reason to assume that there are an unreasonable number of characters to learn. Once a certain number of basic characters are learned, that knowledge can be used in learning new *kanji* words. Miyashita (2004), for instance, shows that the character 止 ('stop') was originally a pictographic symbol depicting a footprint, and that the meanings of other characters containing the element 止 are related to "foot" or 'foot movement"; e.g., 足 ('leg'; foot with the kneecap on the top), 歩 ('walk'), 走 ('run'), 正 ('correct'; to proceed to the person who does not obey and correct him), 歴 ('passage/movement of time'), and 武 ('warrior; to move forward with a weapon'). This type of application of the knowledge of known characters establishes connections between familiar and novel characters.

3. Myth 2: *Kanji* Makes Japanese Reading Difficult

The fact that *kanji* represents both meaning and sound has inevitable consequences for its restricted linguistic functions, visual complexity and similarity, multiple readings, and the large number of characters necessary to represent the concepts that exist in a language (Just and Carpenter 1987, Coulmas 1992). These demanding features of *kanji*, however, can be an advantage for information processing, learning to read, and cultural transmission.

First, *kanji* helps the reader locate crucial information in a text because such information is usually expressed by *kanji* words (Kaiho 1984, Halpern 1990, Shibatani 1990). In addition, the visual complexity of many *kanji* characters makes important information stand out, as compared to *kana,* which tends to have far fewer strokes than *kanji*. The visual density also helps to segment into meaningful units a Japanese sentence, in which words are not usually separated out by spaces as in English. Yamada *et al.* (1991) found that a passage written in *kana* only interfered with native readers' comprehension, suggesting that the combined usage of *kanji* and *kana* facilitates text processing. Furthermore, single- or two-character words convey a great amount of

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

information in a limited space, compared to phonetic transcriptions in *kana* or alphabet letters.

Learning to read in English is not necessarily easier than learning to read in Japanese. Makita (1968), for example, attributed the rarity of reading disabilities among Japanese native children to the logographic nature of *kanji*, based on the observation that English-speaking children with reading disabilities were able to learn *kanji* characters. Though this account is too simplistic, learning to read in English is in fact a challenging task for young children because it involves letter and sound relationships that are frequently irregular. Children's awareness of the sound structure of the language is vital to the acquisition of reading skills in alphabetic languages (e.g., Goswami and Bryant 1990, Wagner *et al.* 1993, McBridge-Chang 1995).

Reading research shows that phonological awareness plays an important role in learning to read in logographic languages as well (e.g., Ho and Bryant 1997, Chow *et al.* 2005, Chan *et al.* 2006, Bialystok and Luk 2007, Fletcher-Flinn *et al.* 2011). Learning to read in Japanese, however, seems to involve different phonological processing skills, which may be attributable to the structural differences between the writing systems (e.g., Mann 1984, 1985; Paradis *et al.* 1985, Yamada *et al.* 1991, Kuhara-Kojima *et al.* 1996, Flaherty 1997). First, there is a relatively strict one-to-one correspondence between *kana* letters and their sounds, and the grapheme-sound mapping is rather straightforward. Second, since *kanji* causes orthographic, semantic, and phonological activations, phonological engagement of neural networks for *kanji* processing is different from that for *kana* or English words (Bolger *et al.* 2005, Tan *et al.* 2005, Perfetti *et al.* 2007).

The logographic nature of *kanji* allows for the orthographic disambiguation of homophonous words, especially those of the Sino-Japanese variety that exist in large numbers due to the simpler phonology and a lack of tones in Japanese as compared to Chinese. The electronic dictionary *Kōjien* (2005), for example, lists over 30 words different in *kanji* that are pronounced /kikoo/ (危行, 気孔, 気功, 気候, 奇功, 奇行, 奇効, 季候, 紀行, 紀綱, 起工, 起稿, 帰向, 帰耕, 帰航, etc.). Obviously, the differences in meaning would be lost if they were written in *kana*.

Lastly, *kanji* knowledge is an invaluable cultural heritage that provides a basis for the understanding of ancient, as well as modern, East Asian culture. Fortunately, as the semantic functions of Chinese characters have been preserved across the Sinosphere (*kanji bunka-ken*)

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

over thousands of years (S. Suzuki 1990), the knowledge of *kanji* not only helps modern-day readers understand the contemporary culture of the Sinosphere but also opens a path to the wealth of knowledge of ancient times, as in classic literature. Haarmann (1990) argues that the microstructure of ideographic symbols reflects the conceptualization of reality that is culturally specific, and that the learning of the compositional structures of *kanji* characters and their components is related to the learning of cultural tradition. *Kanji* learning, therefore, offers an interface between language and cultural learning, and the acquisition of extensive *kanji* knowledge makes both historical and contemporary documents accessible to current and future readers.

Obviously, L2 readers with insufficient knowledge may not be able to take full advantage of these valuable features of *kanji*. As a result, the perception of difficulty may persist as a result of unsuccessful learning. Advanced L2 Japanese students, however, express a lower degree of perceptual difficulty than do less proficient learners (Mori 1999a), suggesting that cumulated learning with an increase in *kanji* knowledge alters learner perceptions, or, alternatively, that a reduced perceptual difficulty facilitates advanced learning. As we have seen, even a small piece of information about a single *kanji* (e.g., 止) can expand one's knowledge about associated characters sharing the same element (足, 歩, 走, 正, 歴, 武 etc.). It is beneficial to employ an explicit strategy instruction in which students are encouraged to connect new information to their existing knowledge.

4. Myth 3: It Is Easy to Guess the Exact Meaning of an Unfamiliar *Kanji* Word Because Its Components Provide Sufficient Information about Word's Meaning

The meanings of novel *kanji* compounds are often inferable because many of them are composed of familiar characters used on a daily basis. Information from word elements, however, must be used with caution since it is not always straightforwardly related to the meaning of the entire word (Mori 2003b). Consequently, L2 Japanese learners often misinterpret unfamiliar words (Mori 2002) or fail to infer at all (Kondo-Brown 2006). The word 外見 (/gaikeN/, 'appearance'), for instance, consists of high frequency characters familiar to most students in an introductory Japanese course: 外 'outside, foreign' and 見 'to see, look.' Students who know the basic meanings of the separate characters may still incorrectly infer the meaning of the compound by combining the

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

semantic contributions of the component characters (e.g., 'looking outside,' 'view of the outside,' 'outside perspective,' and 'foreign sightseeing'). Similarly, the composite character 婿 'son-in-law; bridegroom' refers to a male in spite of the fact that it contains the radical 女 'woman.'

This phenomenon is known as the semantic semitransparency of compounds. That is, *kanji* words often consist of high-frequency familiar elements or characters, but the meaning of an entire word cannot necessarily be obtained from the combined meanings of its individual parts or characters. There are several reasons for this. First, many *kanji* characters or radicals are polysemous, and the identical component can represent different meanings in different words (e.g., 'hand' or 'person' as in 手話 /syuwa/ 'sign language' and 運転手 /uNteNsyu/ 'driver'). Second, semantic information given by word elements is often only partially related to the meaning of the entire word. The 'water' radical 氵, for example, appears in many characters (e.g., 海 'ocean,' 池 'pond,' 流 'current'), but its prototypical meaning is not always relatable to the sense of the entire character unless some etymological explanations are available; e.g., 決 'decision,', 泊 'stay,' 治 'peace.'

Generally speaking, high-frequency characters or radicals tend to have a broader range of semantic functions than low-frequency counterparts because they are used in more contexts; e.g., the water radical in 流, 清, 治, 活, 渋. As a result, words consisting of high-frequency elements tend to be less semantically transparent than those of low-frequency ones. In addition, the meanings of word components are often used metaphorically or figuratively, and such usage is largely culturally bound. The word 青春 /seisyuN/ 'youth,' for instance, only makes sense to those who know that the color blue/green symbolizes the notion of youth, and that the spring season is associated with being hopeful.

This situation is certainly not limited to the learning of *kanji* words. Morphologically complex English words formed by suffixation and compounding are not always morphologically transparent, either. Students should be encouraged to consult additional sources of information to clarify the meaning that is partially determined by word components (Nagy and Anderson 1984, Anderson and Nagy 1991). The combination of the meanings of the two characters 外 'outside, foreign' and 見 'to see, look,' for instance, would yield a number of incorrect guesses, as we have seen earlier, but even the short sentence 人は外見だ

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

けではどんな人かわからない "You cannot tell a person only from his/her outside/foreign + to see/look" can substantially narrow down the candidates for the meaning of the target word. Research findings show that L2 Japanese students have the ability to integrate morphological and contextual information in interpreting novel *kanji* compounds consisting of familiar characters (Mori and Nagy 1999; Mori 2002, 2003b; Mori *et al.* 2007).

In short, learning individual characters and/or their subparts (radicals) is important, but the knowledge of isolated characters/elements does not guarantee the accurate understanding of novel *kanji* words. Hence, L2 readers should keep in mind that overreliance on a single source of information will likely yield inaccurate interpretations, and that when interpreting new *kanji* words they may need to consult multiple sources of information; e.g., semantic or phonetic radicals, component characters, sentential and discourse contexts, word glosses, dictionaries, the content, and world knowledge.

5. Myth 4: Because the Primary Function of *Kanji* Is to Represent Meaning, Sound Plays an Unimportant Role in *Kanji* Recognition

The critical difference between *kanji* and *kana* lies in their basic unit of representation. Each *kanji* character is associated with both meaning(s) and sound(s), whereas a *kana* character represents a sound, or a mora (Otake *et al.* 1993). (Exceptions are the small や *ya*, ゆ *yu*, and よ *yo*.) Since the primary function of *kanji* is to represent meaning, it may be assumed that a *kanji* character would be directly associated with its meaning without phonological mediation, though *kana* processing always involves phonological activation prior to semantic activation. The findings of experimental research, however, suggest that this view may be too simplistic.

The role of phonological recoding during word recognition and silent reading is a major concern among reading researchers. One reason for this is that the capacity of working memory for acoustically encoded stimuli is greater than that for visually encoded input (Gathercole and Baddeley 1993). Since reading comprehension involves the use of working memory to interpret incoming information from a text, the importance of phonological processing of printed materials has become an issue in reading research (e.g., Stanovich 1991, Segalowitz and Hebert 1990).

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

Most experimental psychologists agree on the universal phonological principle that readers, regardless of their language backgrounds, use phonological codes to store information in working memory. Phonological similarities in *kanji* words, for instance, cause experienced readers more confusion than do visual similarities, suggesting that phonological activation does occur during *kanji* recognition (Perfetti and Zhang 1991, Cheng 1992, Wydell *et al.* 1993, Zhang and Perfetti 1993, Wydell *et al.* 1995, Flaherty and Moran 1999, Matsunaga 1999, Morita and Matsuda 2000). In Mori's (1998) study of L2 Japanese students with an alphabetic language background, memory was more interfered with by characters which participants did not know how to pronounce than by those they did.

There has not been consensus among researchers, however, as to the locus of phonological activation during word recognition (Kess and Miyamoto 1997). Within the framework of connectionism (McClelland and Rumelhart 1981), Seidenberg and his colleagues (Seidenberg 1985, Seidenberg and McClelland 1989) have proposed the "time course model," in which phonological and graphemic activations occur in parallel sequence. In this model, the rate at which information becomes available is determined by the weight on connections between units (i.e., orthographic, phonemic, and semantic units), and the locus of phonological activation varies depending on the frequency of words. Another camp of researchers argues that phonological recoding is an automatic process that cannot be easily suppressed, and that it occurs before semantic access regardless of the frequency of words (e.g., Van Orden 1987, 1991, Van Orden *et al.* 1988). Studies of non-alphabetic languages support the time course model (Seidenberg 1985, Besner and Hildebrandt 1987, Flores d'Arcais 1992), as well as the stronger position on phonological activation (Perfetti and Zhang 1991, Cheng 1992, Wydell *et al.* 1993, Perfetti 1995, Wydell *et al.* 1995).

The recent advances in neuroscience technology that enable us to locate brain regions activated during word recognition illuminate both commonalities and variations in neural networks for phonological activation across different orthographic systems. A number of neuroimaging studies have located commonly activated regions across writing systems, as well as unique activations elicited by logographic (Chinese character) processing (e.g., Uchida *et al.* 1999, Tan *et al.* 2001, Thuy *et al.* 2004, Bolger *et al.* 2005, Tan *et al.* 2005, Tham *et al.* 2005, Perfetti *et al.* 2007, Coderre *et al.* 2008, Buchweitz *et al.* 2009). Meta-

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

analyses of neuroimaging studies comparing Asian and alphabetic languages (Bolger *et al.* 2005, Tan *et al.* 2005, Perfetti *et al.* 2007) found strikingly consistent localization across languages in the region known as "the visual word form area" (Cohen *et al.* 2000), suggesting that this area is a gateway region for visual analyses of any words. Chinese character reading showed bilateral activation within the network suggesting that phonological retrieval from characters is cognitively more demanding than that from alphabet or *kana*. Character reading may involve rigorous visuo-spatial analysis demanded by the character's shape-based configuration, which requires collaboration of a number of brain regions in both hemispheres, as opposed to sequential analyses of a string of alphabet or *kana* letters (Tan *et al.* 2001, Thuy 2004).

L2 word recognition is associated with an increased cognitive demand for phonological as well as semantic retrieval and verbal working memory (Wang *et al.* 2003, Liu *et al.* 2006, Liu *et al.* 2007, Perfetti *et al.* 2007, Buchweitz *et al.* 2009). Furthermore, brain activation patterns are influenced by L1-L2 typological differences (e.g., Jeong 2007) and L2 proficiency (e.g., Lee *et al.* 2003, Meschyan and Hernandez 2006). English-speaking college students of L2 Chinese, for instance, acquired the orthographic structure of characters (i.e., stroke and radical analysis) by the end of the first semester, but their lexical access (retrieval of semantic and phonological information) was still slow and difficult in the second term (Wang *et al.* 2003, Liu *et al.* 2006, Liu *et al.* 2007). The lag between the visual (orthographic) analysis and lexical access observed in L2 Chinese was comparable to the character-processing pattern of Chinese natives, suggesting that alphabetic readers develop a neural network that accommodates the demand of the Chinese writing system while learning L2 Chinese. Interestingly, competent Chinese-English bilinguals did not show the alphabetic (left hemisphere dominant) pattern for L2 English, suggesting that a neural system developed for reading Chinese may absorb an alphabetic system (Perfetti *et al.* 2007).

If sound plays an important role in *kanji* word recognition and working memory during reading, and if phonological retrieval of L2 *kanji* is a cognitively demanding skill that requires time and practice to acquire, then the phonetic information of the *kanji* introduced should receive instructional attention. The majority of characters in fact evoke phonological properties, as over 60% of *jōyō kanji* (characters for common use) comprises semasio-phonetic characters (*keisei moji*)—

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

characters consisting of semantic and phonetic elements (Miyashita 2004). Furthermore, the acquisition of efficient, accurate *kanji* naming skills becomes increasingly important in advanced studies not only for reading comprehension but also for phonetic input for *kana*-to-*kanji* conversion in computer inputting (Chikamatsu 2003). Based on the findings of their studies on Chinese readers with limited literacy skills, Chan *et al.* (2006) recommend that character instruction address at least three components of reading-related cognitive skills: learning the orthographic (positional, semantic, and phonological) regularities of characters, enhancing phonological processing skills, and training the rapid naming of characters. As they have obtained positive preliminary results, L2 Japanese students may also benefit from training in these skills.

6. Myth 5: Rote Memorization Is the Only Way to Learn *Kanji*

Rote learning strategies, including repeated writing (Naka and Naoi 1995, Naka 1998), tracking and copying (Onose 1987, 1988), and learning formulaic chunks (e.g. I'm gonna…, I don't know…, …ちゃった) (Myles *et al.* 1998) are commonly used in Japanese schools, and have been found to be effective in learning *kanji* (Onose 1987, 1988, Naka and Naoi 1995, Naka 1998). From the perspective of information processing, repeated practice is indispensable in the acquisition of automatic, efficient lower-level processing skills (e.g., character and/or word recognition), as repeated activations can transform controlled processing that requires much of the learner's attention to automatic processes that require only minimal control (e.g., Shiffrin and Schneider 1977, Anderson 1985).

Repeated writing, however, is not the only way to increase one's character knowledge. *Kanji* knowledge is multi-faceted: students must acquire various types of knowledge even for a single *kanji* or word, including meaning(s), sound(s), orthographic features, compositional structures, stroke order, semantic and/or phonetic congruence with context, grammatical function, and prototypical and/or non-prototypical usage (Kano *et al.* 1993). Learners possess only partial knowledge in each area, but such knowledge is gradually expanded and deepened through multiple exposures (Mori 2003a). Students are thus encouraged to use various learning strategies depending on learning goals and the aspects of *kanji* knowledge they would like to enrich (Mori *et al.* 2007).

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

In their survey of L2 Japanese students' beliefs about *kanji*, Mori and Shimizu (2007) identified six self-reported learning strategies: morphological analysis, rote memorization, context-based strategies, association methods, metacognitive strategies, and "helplessness" (i.e. no strategy). "Morphological analysis" refers to an analysis of parts constituting a *kanji* word. Japanese *kanji* words, as in Chinese, involve two levels of internal structure. One level involves the composition of single-character words consisting of a set of less complex parts, which provide semantic and/or phonetic information that, at least partially, contributes to the meaning and/or the sound of the character. The other level of internal structure involves the composition of *kanji* compounds consisting of two or more characters. In many studies, L2 Japanese students have demonstrated the ability to use their knowledge of the semantic and/or phonetic function of familiar elements/characters while learning new words (Mori and Nagy 1999; Toyoda 1998, 2000; Yamashita and Maru 2000; Kubota and Toyoda 2001; Mori 2002, 2003b; Kondo-Brown 2006; Mori *et al.* 2007).

"Context-based strategies" concern learning new words in context, as opposed to learning words in isolation. L2 learners' ability to use contextual information to learn novel words has been extensively documented (e.g., Day *et al.* 1991, Mondria and Wit-de Boer 1991, Huckin and Bloch 1993, Dubin and Olshtain 1993, Dupuy and Krashen 1993, Fraser 1999, Brown and Sagers 1999, Wode 1999).

"Association methods," or mnemonics, are memory-enhancing strategies using personal encoding techniques or mental images of the items to be learned. Mnemonic devices include keywords (Wang *et al.* 1992, Thomas and Wang 1996, Lu *et al.* 1999), visual images (Wang and Thomas 1992), and phonological resemblance to the target items. Memory aids can be either commonly established or idiosyncratically self-generated (Thomas and Wang 1996, Kuo and Hooper 2004). It is well documented that mnemonics can be beneficially applied to L2 learning in general (Atkinson 1975, Atkinson and Raugh 1975), L2 vocabulary learning (Nation 1990, Van Hell and Candia Mahn 1997, Hopkins and Bean 1998), and short-term retention of Chinese characters (Wang and Thomas 1992, Wang *et al.* 1992, Thomas and Wang 1996). Mnemonics also have long-term benefits for *kanji* learning (Gruneberg 1998, Lu *et al.* 1999).

"Metacognitive strategies" are strategies that are used to become aware of one's own learning process (O'Malley and Chamot 1990,

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

Oxford 1990, Wenden 1998, 1999). Metacognitive knowledge consists of two components: (a) knowledge of cognition, that is, what learners know about their cognitive resources, and (b) regulation of cognition or executional control of the resources (Flavell 1978, Bialystok and Ryan 1985). Metacognitive strategies applied to *kanji* learning include understanding of a given task, selection of suitable *kanji* learning strategies, coordination of one's own learning in an organized manner, monitoring progress, and self-direction in the development of *kanji* knowledge. The identification of "helplessness" as a strategy factor is consistent with the observation teachers often make that students feel lost not knowing effective *kanji* learning strategies (Toyoda 1995, Okita 1997, Gamage 2003). These students will most likely benefit from metacognitive instruction to familiarize them with various learning strategies.

Most importantly, L2 students should be able to use different strategies flexibly and effectively, depending on the nature of a task, and integrate multiple sources of information in order to obtain a better understanding of a new word, rather than from a single source alone (De Bot *et al.* 1997, Mori and Nagy 1999, Paribakht and Wesche 1999, Mori 2002). Unfortunately, Mori and Shimizu (2007) found that L2 Japanese students consider rote memorization the most effective and metacognitive strategies the least effective, suggesting that many do not monitor their own use of the strategies and fail to use multiple sources of information. Incidentally, some L2 Japanese educators consider rote learning the most effective teaching strategy (Shimizu and Green 2002), and student perception may reflect the impact of teacher beliefs or the way that *kanji* is taught. Alternatively, students may have independently decided that rote memorization works best for them simply because they do not know how to use other strategies effectively. While there is nothing wrong with rote learning strategies, which have in fact been found to be effective in *kanji* learning (Onose 1987, 1988, Naka and Naoi 1995, Naka 1998), the lack of self-monitoring and the lack of flexibility in strategy use is a matter of concern.

7. Concluding Remarks

In an attempt to dispel common misconceptions about *kanji*, this article has reviewed the validity of five common *kanji* beliefs often held by L2 Japanese learners. Hopefully, it has succeeded in showing that such folk beliefs are not empirically supported, as they may even promote

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

unwarranted negative attitudes. *Kanji* learning certainly involves much time and effort. Yet, positive feelings and high motivation can enhance L2 learning (e.g., Gardner and Lambert 1972, Dörnyei 2002, Schumann 1994, Ganguly 1994, MacIntyre 2002, Lau and Chan 2003, Ryu Yang 2003). Conversely, negative attitudes toward *kanji* learning and the lack of confidence in one's own learning ability are associated with difficulty in handling novel *kanji* words (Mori 1999b, 2002, Mori *et al.* 2007). Efforts to alert students to the significance of *kanji* from linguistic, cultural, and educational perspectives, and to instill constructive learning attitudes are important as they are bound to increase the chances of students becoming strategic *kanji* learners.

The facts about the number of characters in actual use and their frequency are informative when deciding the number of characters students are expected to learn and help students set achievable goals. Furthermore, discussions of the advantages as well as disadvantages of *kana* and *kanji*, and a better understanding of the usefulness and cultural values of *kanji*, may enable students to view the significance of *kanji* from new perspectives and help them redefine their own *kanji* learning. The importance of using multiple sources of information and strategies while learning new *kanji* words should be emphasized, as overreliance on a single source of information increases the chance of misunderstanding.

At the same time, educators need to deepen students' interests in and promote positive attitudes toward *kanji* by introducing a variety of intellectually stimulating learning activities and providing instructional and moral support to struggling students. Noguchi (2002) reported that her *kanji* learning experience in an introductory course at a U.S. college was disappointing despite the instructor's effort, because characters were taught unsystematically, which made her feel that she had no foundation for learning *kanji.* Later, *Remembering the Kanji* (Heisig 2007), in which characters are presented based on the principles of component analysis (i.e., semantic analysis of elements constituting a character), sparked her interests and helped her to acquire much knowledge of written vocabulary. Lessons of Noguchi's experience are that instruction that heavily emphasizes rote memorization may not work for mature learners with high cognitive capabilities, and that students need to be taught systematically, and understand the reasons for instructional methods used by their teachers.

L2 Japanese students reflect upon their own *kanji* learning from

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

different perspectives and hold different views on the effectiveness of learning strategies. Language educators, therefore, need to listen to what students say about *kanji* learning, identify their *kanji* problem(s), and play a facilitative role by providing effective, achievable scaffolding activities. Such collaboration between students and teachers can create a supportive learning environment and promote student interest in and constructive attitudes toward challenging learning tasks.

An examination of student perceptions is a promising area that can provide explanations of individual differences in strategy use and results of learning. Perception studies have shown that individuals' choice of strategies for a challenging task reflects their views on L2 learning in general (e.g., Horwitz 1988, Oxford 1990, Elbaum *et al.* 1993, Kern 1995, Samimy and Lee 1997, Wenden 1998, Benson and Lor 1999, Cotterall 1995, 1999, Mori 1999b, Sakui and Gaies 1999, Yang 1999), and the nature of a given task in particular (Sanaoui 1995, Gu and Johnson 1996, Parry 1997, Kojic-Sabo and Lightbown 1999, Mori 2002, Mori *et al.* 2007). Future research, therefore, should examine the causal relationship between learner perceptions, learning strategies, and achievement.

NOTE

[1] This article is based on talks the author delivered in the Department of East Asian Studies at Princeton University in March 2000 and at the Princeton in Ishikawa Program in Kanazawa in August 2000. Preparation for the talks and feedback from the audience inspired her to conduct subsequent collaborative research on Japanese language students' perceptions about *kanji* learning (Mori, Sato, and Shimizu, 2007; Mori and Shimizu, 2007). The author sincerely thanks Prof. Seiichi Makino and Prof. Fumiko Nazikian for providing the valuable opportunity.

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

REFERENCES

Anderson, John. 1985. *Cognitive Psychology and Its Implications*, second edition. New York: Freeman.

Anderson, Richard C., and William E. Nagy. 1991. Word Meanings. *Handbook of Reading Research*, ed. by Rebecca Barr, Michael L. Kamil, Peter B. Mosenthal, and P. David Pearson, vol. 2, 690–724. New York: Longman.

Atkinson, Richard C. 1975. Mnemotechnics in Second Language Learning. *American Psychologist* 30: 821–28.

Atkinson, Richard C., and Michael R. Raugh. 1975. An Application of the Mnemonic Keyword Method to the Acquisition of a Russian Vocabulary. *Journal of Experimental Psychology: Human Learning and Memory* 104: 126–33.

Atsuji, Tetsuji. 1993. *Kanji no bekutoru* (The vectors of *kanji*). Tokyo: Chikuma Library.

Benson, Phil, and Winnie Lor. 1999. Conceptions of Language and Language Learning. *System* 27: 459–472.

Besner, Derek, and Nancy Hildebrandt. 1987. Orthographic and Phonological Codes in the Oral Reading of Japanese *Kana*. *Journal of Experimental Psychology: Learning, Memory, and Cognition* 13: 335–43.

Bialystok, Ellen, and Gigi Luk. 2007. The Universality of Symbolic Representation for Reading in Asian and Alphabetic Languages. *Bilingualism: Language and Cognition* 10: 121–129.

Bialystok, Ellen, and Ellen B. Ryan. 1985. A Metacognitive Framework for the Development of First and Second Language Skills. *Metacognition, Cognition, and Human Performance, vol. 1: Theoretical Perspectives*, ed. by Donna-Lynn Forrest-Pressley, G. E. MacKinnon, and T. Gary Waller, 207–52. Orlando: Academic Press.

Bolger, Donald J., Charles A. Perfetti, and Walter Schneider. 2005. Cross-Cultural Effect on the Brain Revisited: Universal Structures Plus Writing System Variation. *Human Brain Mapping* 25: 92–104.

Brown, Cheryl, and Sherri L. Sagers. 1999. Incidental Vocabulary Acquisition from Oral and Written Dialogue Journals. *Studies in Second Language Acquisition* 19: 259–83.

Buchweitz, Augusto, Robert A. Mason, Mihoko Hasegawa, and Marcel A. Just. 2009. Japanese and English Sentence Reading Comprehension and Writing Systems: An fMRI Study of First and Second Language Effects on Brain Activation. *Bilingualism: Language and Cognition* 12: 141–151.

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

Chan, David W., Connie S.-H. Ho, Suk-Man Tsang, Suk-Han Lee, and Kevin K. H. Chung. 2006. Exploring the Reading-Writing Connection in Chinese Children with Dyslexia in Hong Kong. *Reading and Writing* 19: 543–561.

Chikamatsu, Nobuko. 2003. The Effects of Computer Use on L2 Japanese Writing. *Foreign Language Annals* 36: 114–127.

Chow, Bonnie Wing-Yin, Catherine McBridge-Chang, and Stephen Burgess. 2005. Phonological Processing Skills and Early Reading Abilities in Hong Kong Chinese Kindergarteners Learning to Read English as a Second Language. *Journal of Educational Psychology* 97: 81–87.

Cheng, Chao-Ming. 1992. Lexical Access in Chinese: Evidence from Automatic Activation of Phonological Information. *Language Processing in Chinese*, ed. by Hsuan-Chih Chen, and Ovid J. L. Tzeng, 67–91. Amsterdam: Elsevier.

Coderre, Emily L., Christopher G. Filippi, Paul A. Newhouse, and Julie A. Dumas. 2008. The Stroop Effect in *Kana* and *Kanji* Scripts in Native Japanese Speakers: An fMRI Study. *Brain & Language* 107: 124–132.

Cohen, Laurent, Stanislas Dehaene, Lionel Naccache, Stéphane Lehéricy, Dehaene-Lambertz Ghislain, Marie-Anne Hénaff, and François Michael. 2000. The Visual Word Form Area: Spatial and Temporal Characterization of an Initial Stage of Reading in Normal Subjects and Posterior Split-Brain Patients. *Brain* 123: 291–307.

Cotterall, Sara. 1995. Readiness for Autonomy: Investigating Learner Beliefs. *System* 23: 195–346.

———. 1999. Key Variables in Language Learning: What Learners Believe about Them? *System* 27: 493–513.

Coulmas, Florian. 1992. *The Writing Systems of the World.* Oxford: Blackwell.

Day, Richard R., Carole Omura, and Motoo Hiramatsu. 1991. Incidental EFL Vocabulary Learning and Reading. *Reading in a Foreign Language* 7: 541–51.

De Bot, Kees, T. Sima Paribakht, and Marjorie B Wesche. 1997. Toward a Lexical Processing Model for the Study of Second Language Vocabulary Acquisition. *Studies in Second Language Acquisition* 19: 309–29.

Dubin, Fraida, and Elite Olshtain. 1993. Predicting Word Meanings from Contextual Clues: Evidence from L1 Readers. *Second Language Reading and Vocabulary Learning,* ed. by Thomas Huckin, Margot Haynes, and James Coady, 181–202. Norwood, N. J.: Ablex.

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

Dörnyei, Zoltán. 2002. The Motivational Basis of Language Learning Tasks. *Individual Differences and Instructed Language Learning*, ed. by John P. Robinson, 138–58. Amsterdam: John Benjamins.

Dupuy, Beatrice, and Stephen D. Krashen. 1993. Incidental Vocabulary Acquisition in French as a Foreign Language. *Applied Language Learning* 4: 55–63.

Elbaum, Batya E., Cynthia. A. Berg, and David H. Dodd. 1993. Previous Learning Experience, Strategy Beliefs, and Task Definition in Self-Regulated Foreign Language Learning. *Contemporary Educational Psychology* 18: 318–36.

Flaherty, Mary. 1997. The Role of Abstract Visual Memory in the Learning of *Kanji* Reading. *Cognitive Processing of Chinese and Related Asian Languages,* ed. by Hsuan-Chih Chen, 389–99. Hong Kong: The Chinese University Press.

Flaherty, Mary, and Aidan Moran. 1999. Acoustic and Visual Confusions in Immediate Memory in Japanese and English Speakers. *Psychologia* 40: 80–88.

Flavell, John H. 1978. Metacognitive Development. *Structural/Process Theories of Complex Human Behavior*, ed. by Joseph M. Scandura and Charles J. Brainerd, 213–45. Alphen aan den Rijn, The Netherlands: Sijthoff and Noordhoff.

Fletcher-Flinn, Claire M., Brian G. Thompson, Megumi Yamada, and Makiko Naka. 2011. The Acquisition of Phoneme Awareness in Children Learning the *Hiragana* Syllabary. *Reading and Writing* 24: 623–633.

Flores d'Arcais, Giovanni B. 1992. Graphemic, Phonological, and Semantic Activation Processes during the Recognition of Chinese Characters. *Language Processing in Chinese*, ed. by Hsuan-Chih Chen, and Ovid J. L. Tzeng, 37–66. Amsterdam: Elsevier.

Fraser, Carol A. 1999. Lexical Processing Strategy Use and Vocabulary Learning through Reading. *Studies in Second Language Acquisition* 21: 225–41.

Gamage, Gayathri H. 2003. Perceptions of *Kanji* Learning Strategies: Do They Differ among Chinese Character and Alphabetic Background Learners? *Australian Review of Applied Linguistics* 26: 17–30.

Ganguly, S. R. 1994. Attitudes in Second Language Learning: A Social Psychological Perspective. *Psycho-Lingua* 24: 15–24.

Gardner, Robert C., and Wallace E. Lambert. 1972. *Attitudes toward Motivation in Second Language Learning.* Rowley, Mass.: Newbury House Publishers.

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

Gathercole, Susan E., and Alan D. Baddeley. 1993. *Working Memory and Language*. Hove, U.K.: Erlbaum.

Goswami, Usha, and Peter Bryant. 1990. *Phonological Skills and Learning to Read*. East Sussex, U.K.: Erlbaum.

Gruneberg, Michael. M. 1998. A Commentary on Criticism of the Keyword Method of Learning Foreign Languages. *Applied Cognitive Psychology* 12: 529–532.

Gu, Yongqi, and Robert K. Johnson. 1996. Vocabulary Learning Strategies and Language Learning Outcomes. *Language Learning* 46: 643–79.

Haarmann, Harald. 1990. *Language in Its Cultural Embedding: Explanations in the Relativity of Signs and Sign Systems*. New York: Mouton de Gruyter.

Hatano, Giyoo, Keiko Kuhara, and Michael Akiyama. 1997. *Kanji* Help Readers of Japanese Infer the Meaning of Unfamiliar Words. *Mind, Culture, and Activity: Seminal Papers from the Laboratory of Comparative Human Cognition*, ed. by Michael Cole, Yrjo Engeström, and Olga Vasquez, 269–78. Cambridge: Cambridge University Press.

Heisig, James. W. 2007. *Remembering the* Kanji*: A Complete Course on How Not to Forget the Meaning and Writing of Japanese Characters*, fifth edition. Honolulu: University of Hawaii Press.

Halpern, Jack. 1990. *NTC's New Japanese-English Character Dictionary*. Lincolnwood, Ill.: National Textbook.

Ho, Connie Suk-Han, and Peter Bryant. 1997. Phonological Skills Are Important in Learning to Read Chinese. *Developmental Psychology* 33: 946–951.

Horwitz, Elaine K. 1988. The Beliefs about Language Learning of Beginning University Foreign Language Students. *The Modern Language Journal* 72: 283–294.

Hopkins, Gary, and Thomas W. Bean. 1998. Vocabulary Learning with the Verbal-Visual Word Association Strategy in a Native American Community. *Journal of Adolescent and Adult Literacy* 42: 274–281.

Huckin, Thomas N., and Joel Bloch. 1993. Strategies for Inferring Word-Meanings in Context: A Cognitive Model. *Second Language Reading and Vocabulary Learning,* edited by Thomas Huckin, Margot Haynes, and James Coady, 153–76. Norwood, N. J.: Ablex.

Jeong, Hyeonjeong, Motoaki Sugiura, Yuko Sassa, Satoru Yokoyama, Kaoru Horie, Shigeru Sato, Masato Taira, and Ryuta Kawashima. 2007. Cross-linguistic Influence on Brain Activation during Second Language Processing: An fMRI Study. *Bilingualism: Language and Cognition* 10: 175–187.

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

Joyce, Terry. 2011. The Significance of the Morphographic Principle for the Classification of Writing Systems. *Written Language & Literacy* 14: 58–81.

Just, Michael A., and Patricia A. Carpenter. 1987. *The Psychology of Reading and Language Comprehension.* Boston: Allyn and Bacon.

Kaiho, Hiroyuki. 1984. *Kanji o kagaku-suru* (Science of *kanji*). Tokyo: Yūhikaku.

Kano, Cheiko, Yuri Shimizu, Hiroko Takenaka, Eriko Ishii, and Satoru Akutsu. 1993. *Intermediate* Kanji *Book,* vol. 1. Tokyo: Bonjinsha.

Kern, Richard C. 1995. Students' and Teachers' Beliefs about Language Learning. *Foreign Language Annals* 28: 71–92.

Kess, Joseph F., and Tadao Miyamoto. 1997. Psycholinguistic Aspects of *Hanji* Processing in Chinese. *Mon-Khmer Studies* 27: 349–359.

Kojic-Sabo, Izabella, and Patsy M. Lightbown. 1999. Students' Approaches to Vocabulary Learning and Their Relationship to Success. *The Modern Language Journal* 83: 176–92.

Kondo-Brown, Kimi. 2006. How Do English L1 Learners of Advanced Japanese Infer Unknown *Kanji* Words in Authentic Texts? *Language Learning* 56: 109–53.

Kōjien. 2005. Fifth edition. Tokyo: Iwanami Shoten.

Kubota, Mariko, and Etsuko Toyoda. 2001. Learning Strategies Employed for Learning Words Written in *Kanji* Versus *Kana*. *Australian Review of Applied Linguistics* 24: 1–16.

Kuhara-Kojima, Keiko, Giyoo Hatano, Hirofumi Saito, and Tomokazu Haebara. 1996. Vocalization Latencies of Skilled and Less Skilled Comprehenders for Words Written in *Hiragana* and *Kanji*. *Reading Research Quarterly* 31: 158–71.

Kuo, Mei-Liang A., and Simon Hooper. 2004. The Effects of Visual and Verbal Coding Mnemonics on Learning Chinese Characters in Computer-Based Instruction. *Educational Technology Research and Development* 52: 23–38.

Lau, Kit-ling, and David W. Chan. 2003. Reading Strategy Use and Motivation Among Chinese Good and Poor Readers in Hong Kong. *Journal of Research in Reading* 26 (2): 177–90.

Lee, Hyung-Suk, Toshikatsu Fujii, Jiro Okuda, Takashi Tsukiura, Atsushi Umetsu, Maki Suzuki, Tatsuo Nagasaka, Shoki Takahashi, and Atsushi Yamadori. 2003. Changes in Brain Activation Patterns Associated with Learning of Korean Words by Japanese: An fMRI Study. *NeuroImage* 20: 1–11.

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

Liu, Ying, Charles A. Perfetti, Min Wang. 2006. Visual Analysis and Lexical Access of Chinese Characters by Chinese as Second Language Readers. *Language and Linguistics* 7: 637–657.

Liu, Ying, Min Wang, Charles A. Perfetti. 2007. Threshold-Style Processing of Chinese Characters for Adult Second-Language Learners. *Memory & Cognition* 35: 471–480.

Lu, Mei-Yan, James M. Webb, David J. Krus, and Laura S. Fox. 1999. Using Order Analytic Instructional Hierarchies of Mnemonics to Facilitate Learning Chinese and Japanese *Kanji* Characters. *The Journal of Experimental Education* 67: 293–311.

MacIntyre, Peter D. 2002. Motivation, Anxiety and Emotion in Second Language Acquisition. *Individual Differences and Instructed Language Learning*, ed. by Peter Robinson, 45–68. Amsterdam: John Benjamins.

Makita, Kiyoshi. 1968. The Rarity of Reading Disability in Japanese Children. *American Journal of Orthopsychiatry* 38: 599–614.

Mann, Virginia A. 1984. Temporary Memory for Linguistic and Nonlinguistic Material in Relation to the Acquisition of Japanese *Kana* and *Kanji*. *Annual Bulletin of the Research Institute of Logopedics and Phoniatrics* 18: 135–151.

———. 1985. A Cross-Linguistic Perspective on the Relation Between Temporary Memory Skills and Early Reading Ability. *Remedial and Special Education* 6 (6): 37–42.

Matsunaga, Sachiko. 1999. Universality in Reading Processes: Evidence from an Eye-Tracking Study. *Psychologia* 42: 290–306.

McBridge-Chang, Catherine. 1995. What Is Phonological Awareness? *Journal of Educational Psychology* 87: 179–192.

McClelland, James L., and David E. Rumelhart. 1981. An Interactive Activation Model of Context Effects in Letter Perception: Part 1. An Account of Basic Findings. *Psychological Review* 88: 375–407.

Meschyan, Gayane, and Arturo E. Hernandez. 2006. Impact of Language Proficiency and Orthographic Transparency on Bilingual Word Reading: An fMRI Investigation. *NeuroImage* 29: 1135–1140.

Miyashita, Hisao. 2004. *Wakere ba mitsukaru shitteru kanji* (*Decomposition will find you familiar characters*). Second edition. Tokyo: Tarō-Jirō-Sha.

Mondria, Jan-Arjen, and Marijke Wit-de Boer. 1991. The Effects of Contextual Richness on the Guessability and the Retention of Words in a Foreign Language. *Applied Linguistics* 12: 249–67.

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

Mori, Yoshiko. 1998. Effects of First Language and Phonological Accessibility on *Kanji* Recognition. *The Modern Language Journal* 82: 69–82.

———. 1999a. Epistemological Beliefs and Language Learning Beliefs: What Do Language Learners Believe about Learning? *Language Learning* 49: 377–415.

———. 1999b. Beliefs about Language Learning and Their Relationship to the Ability to Integrate Information from Word Parts and Context in Interpreting Novel *Kanji* Words. *The Modern Language Journal* 83: 534–547.

———. 2002. Individual Differences in the Integration of Information from Context and Word Parts in Interpreting Unknown *Kanji* Words. *Applied Psycholinguistics* 23: 375–97.

———. 2003a. Vocabulary Acquisition in Japanese. *Daini gengo kenkyū e no shōtai* (Invitation to second language acquisition research), ed. by Yukiko Hatasa, 47–66 in Japanese, 171–86 in English. Tokyo: Kurosio Syuppan.

———. 2003b. The Roles of Context and Word Morphology in Learning New *Kanji* Words. *The Modern Language Journal* 87: 404–20.

Mori, Yoshiko and William Nagy. 1999. Integration of Information from Context and Word Elements in Interpreting Novel *Kanji* Compounds. *Reading Research Quarterly* 34: 80–101.

Mori Yoshiko, Kumi Sato, and Hideko Shimizu. 2007. Japanese Language Students' Perceptions on *Kanji* Learning and Their Relationship to Novel *Kanji* Word Learning Ability. *Language Learning* 57: 57–85.

Mori Yoshiko, and Hideko Shimizu. 2007. Japanese Language Students' Attitudes toward *Kanji* and Their Perceptions on *Kanji* Learning Strategies. *Foreign Language Annals* 40: 472–90.

Morita, Aiko, and Fumiko Matsuda. 2000. Phonological and Semantic Activation in Reading Two-*Kanji* Compound Words. *Applied Psycholinguistics* 21: 487–503.

Myles, Florence, Janet Hooper, and Rosamond Mitchell. 1998. Rote or Rule? Exploring the Role of Formulaic Language in Classroom Foreign Language Learning. *Language Learning* 48: 323–364.

Nagy, William E., and Richard C. Anderson. 1984. How Many Words Are There in Printed School English? *Reading Research Quarterly* 19: 304–330.

Naka, Makiko. 1998. Repeated Writing Facilitates Children's Memory for Pseudocharacter and Foreign Letters. *Memory and Cognition* 26: 804–809.

Naka, Makiko, and Hiroshi Naoi. 1995. The Effect of Repeated Writing on Memory. *Memory and Cognition* 23: 201–12.

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

Nation, Paul. 1990. *Teaching and Learning Vocabulary*. Boston: Heinle and Heinle.

Nation, Paul, and Robert Waring. 1997. Vocabulary Size, Text Coverage, and Word Lists. *Vocabulary: Description, Acquisition and Pedagogy,* ed. by Norbert Schmitt and Michael McCarthy, 6–19. Cambridge, U.K.: Cambridge University Press.

National Language Research Institute. 1963. *Vocabulary and Chinese Characters Ninety Magazines of Today. vol. II: Chinese Characters Frequency Tables*. Tokyo: Shūei Shuppan.

National Language Research Institute. 1971. *Denshikeisanki ni yoru shinbun no goi chōsa (II)* (Survey of vocabulary of modern newspapers, II). Tokyo: Shūei Shuppan.

Noguchi, Mary. 2002. Kanji no Kōka-teki-na Oshiekata: Kōsei-yōso Bunseki (Effective *kanji* teaching method: Component analysis). Paper presented at the Annual Conference of the Canadian Association for Japanese Language Education (CAJLE) in Toronto, Canada. http://www.kanjiclinic.com/canadaforsite.htm, accessed March 14, 2012.

O'Malley, Michael J., and Uhl A. Chamot. 1990. *Learning Strategies in Second Language Acquisition.* New York: Cambridge University Press.

Okita, Yoko. 1997. Students' Beliefs about Learning Japanese Orthography: Beyond the Textbooks. *New Trends and Issued in Teaching Japanese Language and Culture*, ed. by Haruko M. Cook, Kyoko Hijirida, and Mildred Tahara, 61–76. Honolulu: Second Language Teaching and Curriculum Center, University of Hawaii.

Onose, Masato. 1987. The Effect of Tracing and Copying Practice on Handwriting Skills of Japanese Letters in Preschool and First Grade Children. *Japanese Journal of Educational Psychology* 35: 9–16.

———. 1988. Effect of the Combination of Tracing and Copying Practices on Handwriting Skills of Japanese Letters in Preschool and First Grade Children. *Japanese Journal of Educational Psychology* 36: 129–34.

Otake, Takashi, Giyoo Hatano, Anne Cutler, and Jacques Mehler. 1993. Mora or Syllable? Speech Segmentation in Japanese. *Journal of Memory and Language* 32, 258–78.

Oxford, Rebecca L. 1990. *Language Learning Strategies: What Every Teacher Should Know.* Boston: Heinle and Heinle.

Packard, Jerome L. 2000. *The Morphology of Chinese*. Cambridge, U.K.: Cambridge University Press.

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

Paradis, Michael, Hiroko Hagiwara, and Nancy Hildebrandt. 1985. *Neurolinguistic Aspects of the Japanese Writing System.* Orlando, Fl.: Academic Press.

Paribakht, Sima T., and Marjorie Wesche. 1999. Reading and 'Incidental' L2 Vocabulary Acquisition: An Introspective Study of Lexical Inferencing. *Studies in Second Language Acquisition* 21: 195–224.

Parry, Kate. 1997. Vocabulary and Comprehension: Two Portraits. *Second Language Vocabulary Acquisition,* ed. by James Coady, and Thomas Huckin, 55–68. Cambridge, U.K.: Cambridge University Press.

Perfetti, Charles A. 1995. Very Early Phonological Activation in Chinese Reading. *Journal of Experimental Psychology: Learning, Memory, and Cognition* 21: 24–33.

Perfetti, Charles A., Ying Liu, Julie Fiez, Jessica Nelson, Donald Bolger, and Li-Hai Tan. 2007. Reading in Two Writing Systems: Accommodation and Assimilation of the Brain's Reading Network. *Bilingualism: Language and Cognition* 10: 131–146.

Perfetti, Charles A., and Sulan Zhang. 1991. Phonological Processes in Reading Chinese Characters. *Journal of Experimental Psychology: Learning, Memory, and Cognition* 17: 633–643.

Ryu Yang, Jean Sook. 2003. Motivational Orientations and Selected Learner Variables of East Asian Language Learners in the United States. *Foreign Language Annals* 36: 44–56.

Sakui, Keiko, and Stephen J. Gaies. 1999. Investigating Japanese Learners' Beliefs about Language Learning. *System* 27: 473–492.

Samimy, Keiko K., and Yo-An Lee. 1997. Beliefs about Language Learning: Perspectives of First-Year Chinese Learners and Their Instructors. *Journal of the Chinese Language Teachers Association* 32: 40–60.

Sanaoui, Razika. 1995. Adult Learners' Approaches to Learning Vocabulary in Second Languages. *The Modern Language Journal* 79: 15–28.

Schumann, John H. 1994. Where Is Cognition? Emotion and Cognition in Second Language Acquisition. *Studies in Second language Acquisition* 16: 231–242.

Segalowitz, Norman, and Martine Hebert. 1990. Phonological Recoding in the First and Second Language Reading of Skilled Bilinguals. *Language Learning* 40: 503–38.

Seidenberg, Mark S. 1985. The Time Course of Phonological Code Activation in Two Writing Systems. *Cognition* 19: 1–30.

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

Seidenberg, Mark S. and James L. McClelland. 1989. A Distributed, Developmental Model of Visual Word Recognition and Naming. *Psychological Review* 96: 523–568.

Shibatani, Masayoshi. 1990. *The Languages of Japan.* Cambridge: Cambridge University Press.

Shiffrin, Richard M., and Walter Schneider. 1977. Controlled and Automatic Human Information Processing II: Perceptual Learning, Automatic, Attending, and a General Theory. *Psychological Review* 84: 127–90.

Shimizu, Hideko, and Kathy E. Green. 2002. Japanese Language Educators' Strategies for and Attitudes toward Teaching *Kanji. The Modern Language Journal* 86: 228–241.

Stanovich, Keith E. 1991. Changing Models of Reading and Reading Acquisition. *Learning to Read: Basic Research and Its Implications*, ed. by Laurence Rieben and Charles H. Perfetti, 19–32. Hillsdale, N.J.: Erlbaum.

Suzuki, Shuuji. 1990. *Kanji jōhōryoku sai-hakken* (Rediscover the information transmission capabilities of *kanji*). Tokyo: Sōtaku-sha.

Suzuki, Takao. 1990. *Nihon-go to gaikoku-go* (Japanese and foreign languages). Tokyo: Iwanami Shoten.

Tan, Li Hai, Angela Laird, Karl Li, and Peter T. Fox. 2005. Neuroanatomical Correlates of Phonological Processing of Chinese Characters and Alphabetic Words: A Meta-Analysis. *Human Brain Mapping* 25: 83–91.

Tan, Li Hai, Ho-Ling Liu, Charles A. Perfetti, John A. Spinks, Peter T. Fox, and Jia-Hong Gau. 2001. The Neural System Underlying Chinese Logographic Reading. *NeuroImage* 13: 836–846.

Tham, Wendy W.P., Susan J. Richard Liow, Jagath C. Rajapakse, Tan Choong Leong, Samuel E. S. Ng, Winston E. H. Lim, and Lynn G. Ho. 2005. Phonological Processing in Chinese-English Bilingual Biscriptals: An fMRI Study. *NeuroImage* 28: 579–587.

Thuy, Dinh Ha Duy, Kayoko Matsuo, Kimihiro Nakamura, Keiichiro Toma, Tatsuhide Oga, Toshiharu Nakai, Hiroshi Shibasaki, and Hidenao Fukuyama. 2004. Implicit and Explicit Processing of *Kanji* and *Kana* Words and Non-words Studies with fMRI. *NeuroImage* 23: 878–889.

Thomas, Margaret. H., and Alvin Y. Wang. 1996. Learning by the Keyword Mnemonic: Looking for Long-Term Benefits. *Journal of Experimental Psychology, Applied* 2: 330–342.

Tollini, Aldo. 1994. The Importance of Form in the Teaching *Kanji. Sekai-no nihongo kyōiku* (Japanese language education around the globe) 4: 107–116.

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

Toyoda, Etsuko. 1995. Kanji gakushū ni tai-suru gakushū-sha no ishiki (Student perception about *kanji* learning). *Nihongo kyōiku* (Journal of Japanese language teaching) 85: 101–113.

———. 1998. Teaching *Kanji* by Focusing on Learners' Development of Graphemic Awareness. *Australian Review of Applied Linguistics* 15: 155–168.

———. 2000. English-Speaking Learners' Use of Component Information in Processing Unfamiliar *Kanji*. *Australian Review of Applied Linguistics* 23: 1–14.

Uchida, Idai, Hideyuki Kikyo, Kyoichi Nakajima, Seiki Konishi, Kensuke Sekihara, and Yasushi Miyashita. 1999. Activation of Lateral Extrastriate Areas during Orthographic Processing of Japanese Characters Studied with fMRI. *NeuroImage* 9: 208–215.

Van Hell, Janet G., and Andrea Candia Mahn. 1997. Keyword Mnemonic Versus Rote Rehearsal: Learning Concrete and Abstract Foreign Words by Experienced and Inexperienced Learners. *Language Learning* 47: 507–546.

Van Orden, Guy C. 1987. A ROWS Is a ROSE: Spelling, Sound, and Reading. *Memory and Cognition* 15: 181–198.

———. 1991. Phonologic Mediation is Fundamental to Reading. *Basic Processing in Reading: Visual Word Recognition*, edited by Derek Besner and Glyn W. Humphreys, 77–103. Hillsdale, N. J.: Erlbaum.

Van Orden, Guy C., James C. Johnston, and Benita L. Hale. 1988. Word Identification in Reading Proceeds from Spelling to Sound to Meaning. *Journal of Experimental Psychology: Learning, Memory, and Cognition* 14: 371–86.

Wagner, Richard K., Joseph K. Torgesen, Pamela Laughon, Karen Simmons, and Carol A. Rashotte. 1993. Development of Young Readers' Phonological Processing Abilities. *Journal of Educational Psychology* 85: 83–103.

Wang, Alvin Y., and Margaret H. Thomas. 1992. The Effect of Imagery-Based Mnemonics on the Long-Term Retention of Chinese Characters. *Language Learning* 42: 359–376.

Wang, Alvin Y., Margaret H. Thomas, and Judith A. Ouellette. 1992. Keyword Mnemonics and Retention of Second Language Vocabulary Words. *Journal of Educational Psychology* 84: 520–528.

Wang, Min, Charles A. Perfetti, Ying Liu. 2003. Alphabetic Readers Quickly Acquire Orthographic Structure in Learning to Read Chinese. *Scientific Studies of Reading* 7: 183–208.

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms

Wenden, Anita L. 1998. Metacognitive Knowledge and Language Learning. *Applied Linguistics* 19: 515–37.

———. 1999. An Introduction to Metacognitive Knowledge and Beliefs in Language Learning: Beyond the Basics. *System* 27: 435–41.

Wode, Henning. 1999. Incidental Vocabulary Acquisition in the Foreign Language Classroom. *Studies in Second Language Acquisition* 19: 243–58.

Wydell, Taeko N., Brian Butterworth, and Karalyn Patterson. 1995. The Inconsistency of Consistency Effects in Reading: The Case of Japanese *Kanji*. *Journal of Experimental Psychology: Learning, Memory, and Cognition* 21: 1155–69.

Wydell, Taeko N., Eric K. Patterson, and Glyn W. Humphreys. 1993. Phonologically Mediated Access to Meaning for *Kanji*: Is a ROWS Still a ROSE in Japanese *Kanji*? *Journal of Experimental Psychology: Learning, Memory, and Cognition* 19: 491–514.

Yamada, Jun, Yasushi Mitarai, and Tatsuhiro Yoshida. 1991. *Kanji* Words Are Easier to Identify than *Katakana* Words. *Psychological Research* 53: 136–41.

Yamashita, Hiroko, and Yukiko Maru. 2000. Compositional Features of *Kanji* for Effective Instruction. *Journal of the Association of Teachers of Japanese* 34: 159–178.

Yang, Nae-Dong. 1999. The Relationship Between EFL Learners' Beliefs and Learning Strategy Use. *System* 27: 515–35.

Zhang, Sulan, and Charles A. Perfetti. 1993. The Tongue-Twister Effect in Reading Chinese. *Journal of Experimental Psychology: Learning, Memory, and Cognition* 19: 1082–93.

This content downloaded from
129.194.79.92 on Thu, 09 Oct 2025 08:17:19 UTC
All use subject to https://about.jstor.org/terms